

Slicing tree / polish expression packing

Slicing floorplans use through-cuts

Pseudocode

- Slicing polish is a postfix stack
- Modules push rectangles; H stacks vertically; V places side by side
- The sketch names the operators so evaluation stays deterministic
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Golden tokens A D H B V C V E V pack to a nine by three bbox inside the ten by eight

Algorithm sketch — try these

```
INPUT: polish tokens (modules + H/V)
OUTPUT: packing (x,y,w,h) per module
stack-eval postfix:
  module → push rect
  H: pop a,b; stack vertically
  V: pop a,b; place side by side
GOLDEN: A D H B V C V E V
bbox 9×3; legal in 10×8
```

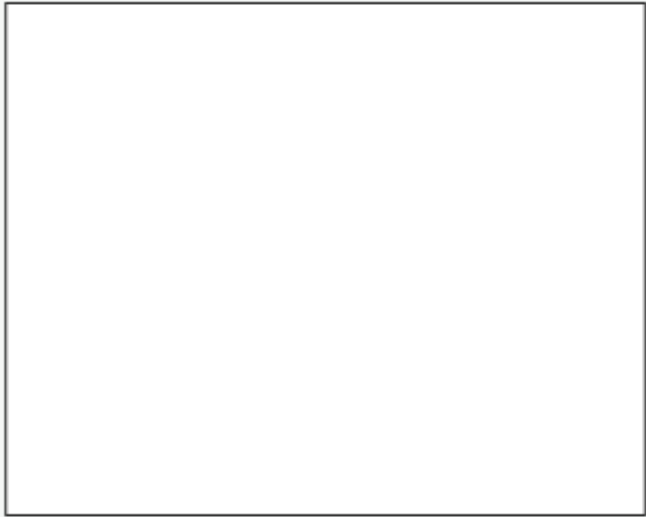
Polish expression encodes cuts

SLICING FLOORPLAN

Polish expression encodes cuts

Step 1 / 5 · polish · module02-01-slicing-floorplan

10×8



Explanation

A slicing floorplan is a tree of through-cuts. Postfix polish A D H B V C V E V stacks A under D, then places B, C, and E to the right with vertical cuts.

- Operands = modules
- H = horizontal cut (stack)
- V = vertical cut (side by side)

```
polish: A D H B V C V E V  
tokens: 9
```

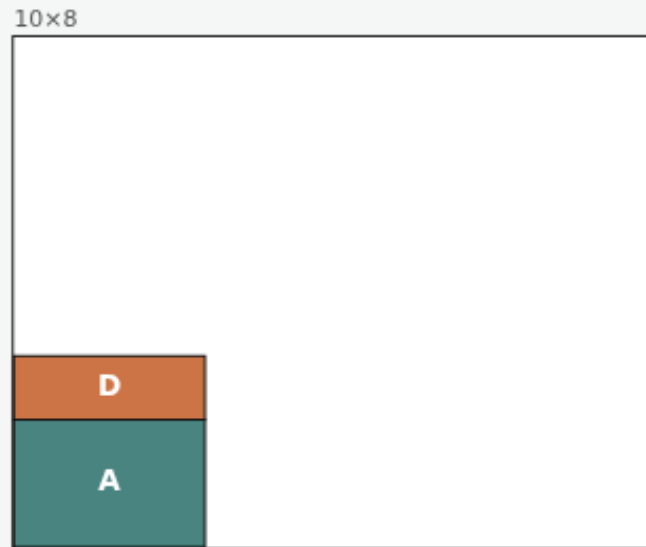
Polish expression encodes cuts

A D H stacks height 3

SLICING FLOORPLAN

A D H stacks height 3

Step 2 / 5 · stack-ad · module02-01-slicing-floorplan



Explanation

A is three by two; D is three by one. An H cut stacks D above A into a three by three block at the lower left.

- A at (0,0)
- D at (0,2)
- composite 3×3

after A D H: w=3 h=3

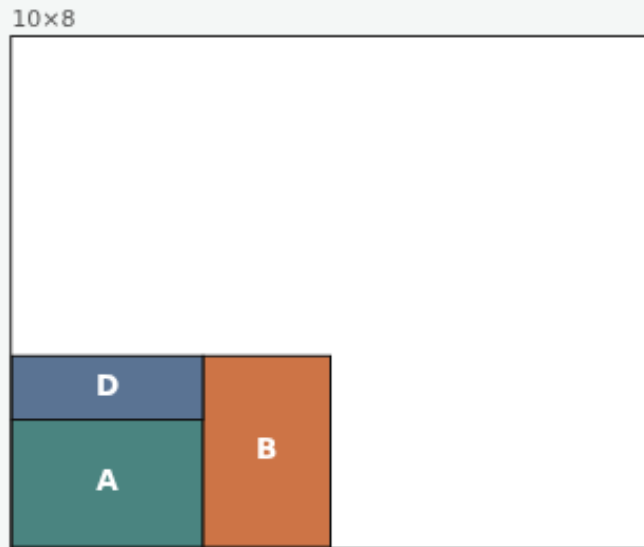
A D H stacks height 3

V attaches B on the right

SLICING FLOORPLAN

V attaches B on the right

Step 3 / 5 · attach-b · module02-01-slicing-floorplan



Explanation

Next V places B beside the A-D stack. Bounding width becomes five; height stays three.

- B at $x=3$
- BB width 5
- height still 3

after ... B V: $w=5$ $h=3$

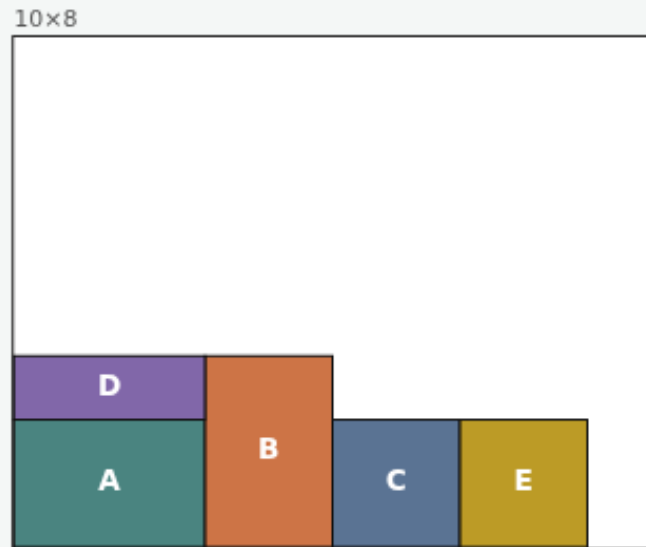
V attaches B on the right

Full polish packs BB 9×3

SLICING FLOORPLAN

Full polish packs BB 9×3

Step 4 / 5 · full · module02-01-slicing-floorplan



Explanation

Adding C and E with more V cuts finishes the golden polish. Bounding box is nine by three—legal inside the ten-by-eight outline.

- Final BB: 9×3
- Fits outline 10×8
- Five modules placed

bb: 9×3
legal: true

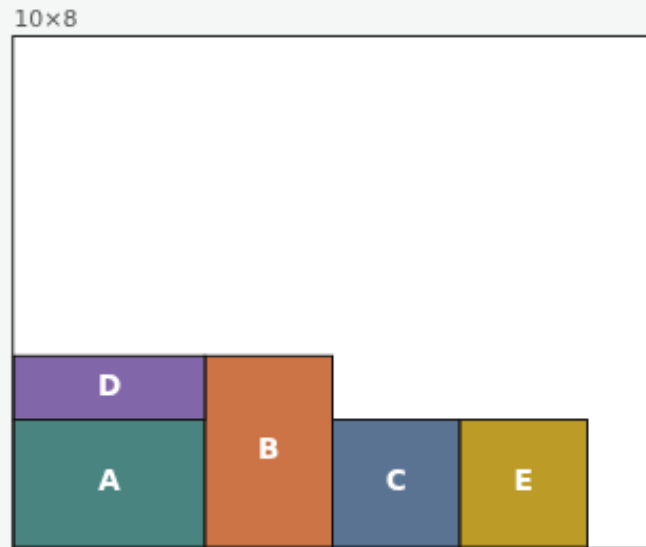
Full polish packs BB 9×3

Slicing cannot make wheels

SLICING FLOORPLAN

Slicing cannot make wheels

Step 5 / 5 · takeaway · module02-01-slicing-floorplan



Explanation

Not every packing is slicing. Wheel topologies need non-slicing codes like B-star or sequence pair—next labs. For slicing, polish plus H/V evaluation is enough.

- Golden polish → BB 9×3
- legal: true
- Non-slicing needs other reps

starter golden: BB 9×3

Slicing cannot make wheels

Browser lab track

- Open slicing-floorplan and Evaluate polish
- Confirm bounding width nine, height three, and a legal packing inside ten by eight

Implement track

- Implement postfix evaluation for H and V
- On the golden token list, assert width nine, height three, and `is_legal_packing` true

Pitfalls

- Swapping H/V meanings

Your turn

- Ship a legal polish pack with BB nine by three
- Next: B-star trees for non-slicing adjacency